CyberRadar

Network Security Monitoring and Analysis Dashboard

Technical Project Report | Draft for project submission

Primary tools: Nmap and Wireshark | Application: Python Flask | Packet parsing: Scapy | Local monitoring: psutil

This report describes the implementation and intended operation of CyberRadar based on the provided project code and feature descriptions. It does not claim independent execution or validation of the application.

1. Abstract

CyberRadar is a locally hosted cybersecurity dashboard that combines Nmap-based localhost port scanning and authorized LAN device discovery with Wireshark-assisted traffic capture and Scapy-based PCAP/PCAPNG analysis. The Flask backend also uses psutil to display live network counters and locally observable process connections. A browser interface presents scan results, device information, protocol statistics and visual charts.

2. Objectives

• Provide a simple interface for inspecting selected localhost TCP port states using Nmap.

• Discover responding hosts on a network the user owns or is authorized to assess.

• Use Wireshark to capture permitted network traffic, then import capture files for Scapy analysis.

• Display laptop network activity and local process connection summaries.

• Explain limitations and safe use of local network security tools.

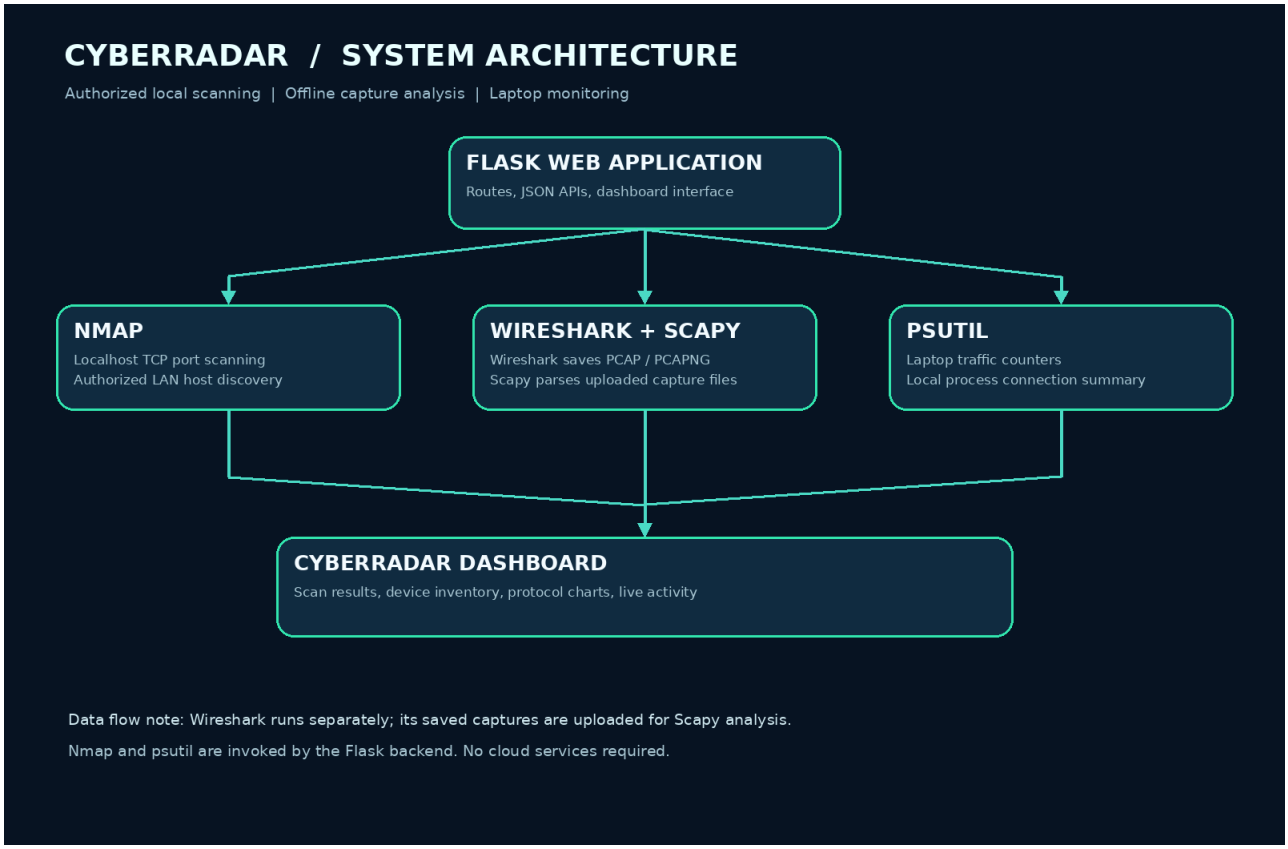
3. Technologies

Technology	Role
Nmap	Port scanning and authorized LAN host discovery
Wireshark	External network capture and packet inspection
Scapy	Parsing uploaded PCAP and PCAPNG captures
Flask / Python	Web routes, scanning orchestration and JSON APIs

Technology	Role
psutil	Laptop network counters and process connections
HTML / CSS / JS / Chart.js	Interactive browser dashboard and charts

4. Architecture and Data Flow

The Flask server exposes routes for the dashboard, local port scan, Wi-Fi discovery, packet upload and live counters. Nmap is invoked locally for scanning. Wireshark runs separately to create packet capture files, which users upload for analysis by Scapy. psutil reads activity from the computer hosting the application. Results are displayed through the frontend.



5. Main Modules

Dashboard: Overview cards, Nmap chart, local scan controls and results. **Home Wi-Fi Scan:** Network information, consent-gated host discovery, device inventory and user-entered labels. **Packet Analyzer:** PCAP/PCAPNG upload, packet counts, protocol distribution, source IP summary and packet table. **Live Monitoring:** Laptop throughput and observable process connections.

6. Typical Usage

1. Start the Flask application and open the local dashboard.

2. Run a localhost Nmap scan and inspect returned port states.

3. Use Wireshark on permitted traffic, save a PCAP or PCAPNG file, then upload it to CyberRadar.

4. On an owned or authorized LAN, confirm permission and run device discovery.

5. Inspect live laptop activity and connection summaries.

7. Testing and Validation

A 12-case test plan is supplied in **docs/testing.md**. It covers UI loading, Nmap scanning, file upload validation, network discovery consent, saved labels and live monitoring. All test outcomes must be completed after execution; this report does not claim measured results.

8. Limitations

• Nmap host discovery can miss devices that block discovery or are isolated from the scanning laptop.

• Device MAC/vendor/hostname information may be missing.

• Localhost port scanning covers selected ports and does not constitute vulnerability assessment.

• Wireshark is not embedded or controlled by Flask; capture files are imported manually.

• Application monitoring only observes the laptop running CyberRadar.

• Device room labels are manual metadata, not physical location tracking.

9. Security and Privacy

CyberRadar should only scan systems and networks with authorization. Do not publish personal packet captures, device inventories, Wi-Fi names, sensitive IP addresses, passwords or credentials. Redact screenshots before sharing publicly and use a .gitignore file to exclude locally generated data.

10. Future Enhancements

Potential improvements include network topology visualization, device-change alerts, historical charts, exportable sanitized reports and broader protocol analysis.

11. Conclusion

CyberRadar demonstrates how Nmap, Wireshark, Scapy and psutil can complement each other in a local educational network security dashboard. The separation between active scanning, external packet capture, offline packet analysis and local monitoring makes the system straightforward to demonstrate and explain.

References / Official Tools

Nmap: nmap.org | Wireshark: wireshark.org | Scapy: scapy.net | Flask: flask.palletsprojects.com | psutil: psutil.readthedocs.io